

Predicting Term Deposit Subscription from Bank Marketing Campaign Data

Villalobos Socualaya, Luis Enrique

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Predicting Term Deposit Subscription from Bank Marketing Campaign Data

Abstract

This study developed and evaluated classification models to predict whether a bank client would subscribe to a term deposit using the UCI Bank Marketing dataset originally documented by Moro et al. (2014). Following the CRISP-DM framework (Shearer, 2000), the analysis used 45,211 observations and applied a leakage-aware preprocessing strategy by excluding call duration, a variable unavailable at decision time. Data preparation included variable renaming, binary encoding, creation of a prior-interaction summary feature, and train-only transformation of model inputs after a stratified 80/20 split.

Five models were compared: a Dummy baseline, Pipeline Logistic Regression, class-balanced Logistic Regression, tuned Random Forest, and tuned XGBoost. Among the evaluated models, tuned XGBoost achieved the best overall balance of minority-class detection and ranking performance, with Accuracy = 0.8014, Precision = 0.3259, Recall = 0.6531, F1-score = 0.4349, and ROC-AUC = 0.8010.

The findings show that accuracy alone is insufficient for this imbalanced marketing problem and that recall-oriented evaluation is more appropriate when the objective is to identify likely subscribers. The final model provides a practical basis for prioritizing customer outreach while also highlighting limitations related to class imbalance, context specificity, and restricted behavioral information.

1. Business Understanding

In the current financial sector, operative efficiency in long term deposits is fundamental for maintaining competitive liquidity levels. Direct marketing campaigns are commonly used by banks to promote financial products and services, such as term deposits. However, these campaigns often involve contacting a large number of customers without knowing in advance who is likely to accept the offer. As a result, organizations may spend considerable time and resources reaching clients who are not interested. In this regard, predictive analytics and machine learning techniques can address this challenge by identifying patterns in customer data and leveraging data-driven insights that can improve marketing efficiency, and support more informed decision-making.

1.1 Problem Definition

Banks frequently rely on telemarketing campaigns to promote their products. However, determining which customers are more likely to subscribe remains a significant challenge. Without predictive tools, marketing teams may contact many clients who decline the offer, leading to inefficient use of resources. This project aims to develop a classification model capable of predicting whether a customer will subscribe to a term deposit based on available customer and campaign information. To address this challenge, the project is guided by the following research question:

How accurately can we predict whether a client will subscribe to a term deposit using customer profile, prior campaign history, and campaign-context information?

1.2 Objectives

The main objective is to develop and evaluate a predictive model that can estimate the probability that a client will subscribe to a term deposit, using customer demographic profiles, financial indicators, and historical campaign data.

Finally, the project seeks to identify key factors associated with successful subscriptions and generate insights that could help improve targeting future marketing campaigns and allocate marketing resources more efficiently. By doing so, financial institutions can optimize resource allocation and increase campaign effectiveness.

1.3 Success criteria

The success of the project will be measured by the ability of the classification model to predict whether customers will subscribe to a term deposit. With that purpose, performance will be evaluated using precision, recall, ROC-AUC, and supporting metrics such as accuracy and F1-score. Given the imbalanced nature of the dataset, particular emphasis is placed on recall and precision. A successful model should demonstrate reliable predictive performance and provide actionable insights that could help financial institutions focus their marketing efforts on customers with a higher likelihood of accepting the offer.

2. Data Understanding

2.1 Dataset Sources and Context

The data was extracted from the UCI Machine Learning Repository, specifically the “Bank Marketing” dataset. It contains data from direct marketing campaigns conducted by a Portuguese banking institution. The objective of this dataset is to predict whether a client will subscribe to a term deposit offered during a telemarketing campaign. This dataset is a recognized benchmark in the field of predictive analytics for business, providing a representation of real-world telemarketing interactions and ensuring relevance of the findings. For the purposes of this analysis, it is assumed that the individuals included in the dataset are existing clients of the bank who were selected as part of the marketing campaign.

2.2 Dataset description

2.2.1 Dataset Dimensions and Data Types

The bank marketing dataset contains 45,211 observations and 17 variables. The dataset includes a combination of 7 numerical, 6 categorical, and 4 binary variables, providing the necessary information to develop a classification model.

2.2.2 Variables description

The data is structured as a classification problem where the target variable (y) indicates whether a client subscribes to a term deposit. The features are organized in four different groups:

Group	Description	Variables
Bank Client Profile	Demographic and financial attributes	-Age -Job -Marital status -Education -Credit in default -Average yearly balance -Housing loan -Personal loan
Current Campaign Data	Attributes related to the last contact of the current campaign	-Contact type -Last contact day -Last contact month -Last call duration -Number of contacts in this campaign
Historical Interaction Data	Indicators of previous interactions	-Number of days passed since the client was last contacted (pdays) for the previous campaign -Number of contacts performed before this campaign.
Outcome Data	Result of the campaigns	-Result of previous campaign -Result of current campaign.

2.2.3 Data Leakage

Further consideration concerns the variable duration. This feature shows a strong correlation with the target; however, as it is only determined after a call is completed, it represents a significant data leakage risk if used for pre-campaign decision making.

2.2.4 Sentinel Values

A key data quality issue involves the variable pdays. This feature represents the number of days since the client was last contacted, but it uses -1 to represent clients who have never been contacted before. Because this value acts as a “sentinel value” rather than a true numerical distance, treating it as a standard numeric value could introduce bias into the model.

2.3 Initial exploration and Data Quality Audit

2.3.1 Data format

An initial inspection of the raw data indicated minor formatting considerations. The dataset was successfully loaded using an appropriate delimiter specification, which ensured that all variables were correctly parsed into their respective columns. This approach allowed the dataset to be structured properly without requiring additional preprocessing steps, preserving the integrity of the original data.

2.3.2 Duplicates and missing values

The dataset was inspected for duplicated records and missing values to ensure data integrity. No duplicated rows were identified in the dataset, confirming that each observation represents a unique client interaction within the campaign.

Regarding missing values, the dataset does not contain explicit null values (NaN); however, four categorical values (job, education, contact, and poutcome) contain the label “unknown”, which represents missing or unrecorded information (Appendix Table 1).

For the variables of job, education, and contact, these entries were retained to evaluate their predictive value in the model. In contrast, for the variable poutcome, the label “unknown” corresponds to clients who were never contacted in a previous campaign.

The application of Little’s MCAR test was considered; however, it was not applied because all variables containing these encoded missing values are categorical, while the test is designed for numerical data.

3. Exploratory Data Analysis (EDA)

This section examines the predictor set before preprocessing and modeling. The goal is to describe the data, assess class balance, compare feature scales, inspect distribution shape, and identify possible multicollinearity. These checks are necessary because later preprocessing choices should be justified by the observed structure of the data rather than applied by default.

3.1 Descriptive Statistics and Feature Scale Comparison

The descriptive statistics show substantial variation across predictors in both central tendency and dispersion. For instance, Balance, Days_Since_Previous_Contact (pdays), and Previous_contacts (previous) have much larger numeric values and wider spreads than binary variables such as Housing_Loan, Personal_Loan, and Credit_Default. For example, Balance has a mean of 1362, a standard deviation of 3044, and a maximum number of 102127, while Housing_Loan has a mean of 0.56, a standard deviation of 0.37, and a maximum number of 1. Full descriptive statistics are reported in Appendix Table 2. These observations highlight the need for further distribution analysis and potential transformations in later stages.

3.2 “Unknown” values check

To assess the impact of “unknown” categories in selected categorical variables (Occupation, Education_Level, Contact_Method, and Previous_Campaign_Outcome), a Chi-square test of independence was conducted against the target variable (Sub_Term_Deposit). This test was applied to determine whether these variables, including their “unknown” categories, are statistically associated with the outcome and therefore potentially informative for prediction.

The results show extremely low p-values across all variables (Appendix, Table 3), indicating a statistically significant association with the target variable. This leads to the rejection of the null hypothesis of independence, confirming that these categorical features contain relevant information for predicting term deposit subscription.

Based on these findings, the “unknown” category was retained as a valid level within each variable, as removing or imputing it could result in the loss of meaningful predictive signals.

3.3 Distribution Shape and Skewness

To assess distribution of numerical values, both visual and statistical methods were applied. Appendix Figure 1 illustrates the shape of each numeric feature, indicating clear deviations from normality. Complementarily, skewness was calculated for each feature using a threshold of $|0.5|$ to identify substantial asymmetry.

Both approaches revealed that some predictors exhibited strong positive skewness, particularly Avg_Yearly_Balance, Campaign_Contacts, Days_since_Previous_Contact, and Previous_Contacts (see Appendix Table 4). Such skewed distributions can negatively impact the performance of certain machine learning algorithms by increasing the influence of extreme values and reducing the stability of model estimates.

Based on these results, appropriate transformations such as Box-Cox or Yeo-Johnson will be analyzed and applied in the data preparation stage to reduce skewness and improve model stability.

3.4 Class Imbalance in Target Variable

A frequency analysis of the target variable reveals a significant imbalance in the data (Appendix Figure 2). Only about 11.7% of the observations correspond to successful subscriptions (“yes”), while the majority represent unsuccessful outcomes (“no”) (Appendix Table 5). This imbalance reinforces the need to apply appropriate techniques to address class imbalance, and accuracy alone will be an insufficient metric for evaluating model performance.

3.5 Outlier analysis

An outlier analysis was conducted on the continuous variables using both the Z-score and Interquartile Range (IQR) method to identify extreme values. The results indicate the presence of a considerable number of outliers in variables such as Avg_Yearly_Balance, Campaign_Contacts, Days_Since_Previous_Contact, and Previous_Contacts (Appendix Table 6). However, these observations were not removed or treated at this stage. Instead, these values were retained because they may reflect real customer behavior rather than data errors and imputing them could introduce bias in the analysis. Additionally, keeping them allows for evaluating the impact of subsequent transformations, and assessing how machine learning models handle these extreme values. This approach ensures that any decision regarding outlier treatment is guided by its impact on model performance rather than solely on statistical criteria.

3.6 Correlation structure and multicollinearity

The correlation analysis indicates weak relationships among the predictors (Appendix Table 7, Figure 3). The strongest correlation is observed between Days_Since_Previous_Contact and Previous_Contact, reflecting a moderate association related to prior campaign interactions. The correlation analysis is conducted after transforming binary variables into a numerical format to ensure that all relevant predictors are included in the analysis. However, since correlation measures linear relationships between numerical variables, categorical variables are not considered at this stage. This approach allows for a more complete assessment of relationships between predictors, particularly those that were originally categorical but are now represented in numeric form.

Furthermore, the exploration revealed a conceptual link between Days_Since_Previous_Contact, Previous_Contact, and Previous_Campaign_Outcome. Clients who were never contacted are typically represented by the combination previous = 0, pdays = 1, and poutcome = “unknown”. This indicates that these variables capture overlapping aspects of prior campaign history, which may introduce multicollinearity in the feature space.

All other variables exhibit substantially lower correlations, with absolute values below 0.19. However, this step remains exploratory in nature and is used to identify potential multicollinearity issues before modeling, rather than to perform feature selection at this stage. For assessing Multicollinearity in the model, the Variance Inflation Factor (VIF) is applied after splitting.

3.7 Key Findings from EDA

The exploratory analysis identifies four fundamental issues for the next stage of the project. First, the predictors operate on vastly different scales, which supports later feature transformations.

Second, several variables, especially `Previous_Contacts`, are strongly right skewed with significant outliers and should be handled carefully during preprocessing and interpretation. Third, there is an important imbalance in the target variable, which indicates the need to implement different methods that can deal with the difference in distribution and bring accurate results. Finally, some categorical variables are conceptually linked, which may introduce potential multicollinearity in the analysis.

4. Data Preparation and preprocessing

4.1 Data cleaning

The data cleaning stage focused on ensuring that the dataset was properly structured and free from inconsistencies that could affect the analysis. In this stage, the variable `duration` was removed prior to model training because it is only known after the call has been completed. Including this variable would introduce data leakage and lead to unrealistic predictive performance.

4.2 Feature Processing

This stage encompasses all steps to prepare variables for modeling, including renaming, encoding, scaling, and feature engineering.

4.2.1 Rename columns

To improve clarity and interpretability, the original variable names were renamed using more descriptive and user-friendly labels. This transformation aimed to make the dataset more accessible for analysis and easier to interpret in a business context, without altering the underlying information. The updated naming convention reflects the meaning of each variable more explicitly, facilitating both model development and communication of results. The detailed change is Appendix Table 8.

4.2.2 Binary Variables Encoding

For modeling purposes, all binary variables in the dataset were encoded into numerical form. This includes both the target variable and selected predictor variables.

The target variable was encoded such that yes (term deposit acceptance) = 1 and no (rejection) = 0, aligning with the evaluation focus of the analysis where identifying successful subscriptions is the primary objective.

Similarly, binary predictor variables such as Credit_Default, Housing_Loan, and Personal_Loan were encoded into 0/1 format to ensure compatibility with machine learning algorithms. This consistent encoding allows the model to interpret variables effectively while preserving their original meaning.

4.2.3 Feature Scaling Approach

As mentioned in the previous section, several numerical variables exhibited substantial differences in scale and skewness. To address this issue, power transformations were applied after the train-test split to prevent data leakage. Initial skewness screening is summarized in Appendix Table 4; however, the final implemented transformations in the modeling workflow were Campaign_Contacts → Box-Cox and Age, Avg_Yearly_Balance, and Last_Contact_Day → Yeo-Johnson. This approach helped stabilize skewed distributions while ensuring that transformation parameters were learned only from the training set and then applied consistently to the test set.

4.2.4 Categorical variables encoding

Categorical variables such as Occupation, Marital_Status, Education_Level, Contact_Method, Last_Contact_Month, and the engineered variable Previous_Interaction_Status were encoded using OneHotEncoder to convert them into numerical representations suitable for model training. The encoding was performed after splitting the data to avoid data leakage.

4.2.5 Feature Engineering

To address redundancy among variables related to prior campaign history, a feature engineering step was applied to `Days_Since_Previous_Contact`, `Previous_Contacts`, and `Previous_Campaign_Outcome`. These variables were consolidated into a single categorical feature, `Previous_Interaction_Status`, which summarizes the client's prior interaction history in a more interpretable way. Following this transformation, the original variables were removed to reduce redundancy and improve feature clarity.

4.3 Multicollinearity Assessment

After applying feature engineering, multicollinearity was evaluated using the Variance Inflation Factor (VIF). The results show a collinearity falling within acceptable thresholds (see Appendix Table 9). The highest VIF value corresponds to `Previous_Interaction_Status_Never_Contacted` (8.85), followed by some dummy variables derive from One-Hot Encoding. These values are expected due to the nature of categorical expansion and do not indicate problematic multicollinearity. In this range, multicollinearity is generally interpreted as moderate rather than severe and therefore does not necessarily compromise model stability or interpretability.

Additionally, many of these values are associated with dummy variables generated through One-Hot Encoding, where higher VIF values are expected due to the inherent structure of categorical expansion. Removing these variables could result in loss of important information and reduce model interpretability.

Because no VIF value exceeded the commonly accepted threshold of 10 and model performance remained stable across evaluation metrics, multicollinearity was not considered severe enough to justify removing additional predictors. This decision ensures a balance between statistical rigor and preserving meaningful predictive features.

5. Modeling

5.1 Train/Test Split Strategy

The predictor matrix was defined using the 13 remaining features after removing duration and consolidating prior-contact variables through feature engineering, and the binary target variable was created from Sub_Term_Deposit, with subscription encoded as the positive class.

The dataset was split into training and test sets using an 80/20 stratified partition, resulting in approximately 36,168 observations for training and 9,043 for testing. The split was performed before transformation or any other train-learned preprocessing step.

5.1.1 Rationale for Stratification

The split was performed using stratification to preserve the class proportions of the original dataset in both subsets. Before and after the split, the target distribution was 11.7% yes and 88.3% no in both the training and testing sets. A summary of class proportions before and after splitting is provided in Appendix Table 10. The consistency of these proportions with the original distribution confirms that stratification worked as intended.

5.2 Model Baseline: *DummyClassifier*

A baseline establishes the minimum level of performance that any substantive classifier must exceed. Without a baseline, later results can appear stronger than they actually are, especially in datasets with class imbalance. In this project, a *DummyClassifier* was used as the initial benchmark, as it represents a naive strategy that predicts only the majority class. By following this approach, the model serves as a necessary reference point to quantitatively measure the impact of the dataset's inherent class imbalance on standard linear algorithms.

On the test set, the baseline achieved an accuracy of 0.8830, but completely failed to identify the minority class, with precision, recall, and F1-score all equal to 0, and a ROC-AUC of 0.50. A full summary of the baseline matrix is provided in Appendix Table 11, and the confusion matrix is shown in Appendix Figure 4.

5.3 Comparison models

The performance of the baseline model illustrates why accuracy cannot be interpreted in isolation. Although the model appears to perform well in terms of overall accuracy, it provides no practical value for the business objective, as it fails to identify any potential subscribers. This baseline therefore establishes a lower bound for model performance. Consequently, these results provide a quantitative justification for transitioning to more robust ensemble methods and implementing cost-sensitive learning techniques, such as class-weight adjustments or ensembled methods, to accurately predict product subscribers.

5.3.1 Model 1: Logistic Regression (Pipeline)

As a substantive model, a Logistic Regression was implemented within a preprocessing pipeline. This approach integrates feature scaling (Yeo-Johnson) and categorical encoding (One-Hot Encoding), ensuring that all transformations are applied in a leakage-safe manner. The pipeline combines both numerical and categorical preprocessing steps using ColumnTransformer. The model uses the ‘liblinear’ solver with an ‘L1’ penalty. Overall, the pipeline provides a clean and modular framework integrates preprocessing and modeling into a consistent process.

5.3.2 Model 2: Logistic Regression Balanced

To address the high false negative rate observed in previous iterations, this model utilizes ‘balanced’ class-weight strategy with ‘liblinear’ solver. This configuration forces the algorithm to penalize the misclassification of subscribers more heavily than non-subscribers, effectively recalibrating the decision boundary to prioritize recall over global accuracy.

5.3.3 Model 3: Random Forest with GridSearch

Random Forest was included to create a more stable and accurate prediction by aggregating multiple trees. As an ensemble method, it improves predictive stability by aggregating multiple decision trees. Hyperparameter tuning was conducted using GridSearchCV, the best cross-validated result was obtained with 300 trees, `max_depth = 8`, `min_samples_leaf = 5` and `class_weight = ‘balanced’`.

5.3.4 Model 4: XGBoost with Imbalance Handling

XGBoost was implemented as an advanced model capable of capturing complex non-linear patterns. To address class imbalance, the parameter `scale_pos_weight` was computed based on the class distribution in the training data.

The hyperparameter tuning grid covered a focused set of parameters related to tree depth, learning rate, subsampling, and column subsampling. This kept the search broad enough to be meaningful while remaining appropriate for the dataset size. The selected configuration used 200 estimators, `max_depth = 3`, and `learning_rate = 0.1`, `subsample = 0.7`, and `colsample_bytree = 0.7`.

5.4 Hyperparameter Tuning Strategy

Hyperparameter tuning was conducted using GridSearchCV on the training data only. The tuning objective was Recall, as the primary business objective is to minimize false negatives (i.e., missed subscribers). Using recall as the optimization metric ensures that model selection is aligned with the error structure of the task rather than driven by accuracy alone.

The tuning grids were intentionally compact. This was a deliberate choice to avoid unnecessary complexity while still allowing meaningful variation in model behavior. The goal at this stage was to identify reasonable configurations for each model rather than to exhaustively search every possible setting.

5.5 Cross-Validation Design

Cross validation was implemented to ensure that model performance is robust and generalizable. All validation procedures were conducted exclusively on the training data to prevent data leakage and preserve the integrity of the test set as a final evaluation benchmark.

6. Model Evaluation and Comparison

6.1 Evaluation Metrics Used

Model evaluation was based on accuracy, precision, recall, F1-score, ROC-AUC, and the confusion matrix. This metric set was selected to reflect both overall predictive and the business importance of subscriber detection. In this context, recall is especially important because a False Negative translates to failing to contact a client who would have accepted the product, leading to a loss in sales. However, recall was not interpreted in isolation; the confusion matrix, along with the precision, F1-Score, and ROC-AUC were used to assess the model's reliability in predicting term deposit subscriptions. This broader evaluation strategy is necessary because a single metric, especially accuracy, can conceal business-critical weaknesses.

6.1.1 Confusion Matrix

The confusion matrices show clear differences in error patterns across models. Logistic regression (pipeline) produced 7,891 True Negatives, 190 True Positives, 94 False Positives, and 868 False Negatives (Appendix, Figure 5). Logistic regression (balanced) obtained 6,207 True Negatives, 676 True Positives, 1,778 False Positives, and 382 False Negatives (Appendix, Figure 6). Random forest produced 6,478 True Negatives, 655 True Positives, 1,507 False Positives, and 403 False Negatives (Appendix, Figure 7). XGBoost produced 6,556 True Negatives, 691 True Positives, 1,429 False Positives, and 367 False Negatives (Appendix, Figure 8).

These results highlight that the models perform differently in terms of error trade-offs. XGBoost produced the lowest number of false negatives, indicating the strongest ability to identify potential subscribers. Logistic Regression (balanced) also performed well in reducing false negatives, while the pipeline version showed significantly weaker performance in this regard.

In contrast, XGBoost and Random Forest produce a higher number of true positives, but at the cost of substantially more false positives. This reflects a more aggressive classification strategy that prioritizes identifying potential subscribers at the expense of contacting more uninterested clients.

In the context of marketing campaigns, this trade-off is particularly important: false negatives represent missed business opportunities, while false positives imply additional operational costs due to unnecessary customer contact. Therefore, models such as XGBoost and Logistic Regression (balanced) may be more suitable when broader customer coverage is desired, even at a higher operational cost.

6.1.2 Accuracy, Precision, Recall, and F1-Score Results

The Logistic Regression (Pipeline) achieved an accuracy superior to the baseline of 0.8936, a Precision 0.6690, Recall 0.1796, and F1-Score 0.2832. While its balance version significantly improves recall (0.6389) but sacrifices precision (0.2755) and accuracy (0.7611).

Random Forest achieved a recall (0.6191) slightly lower than the balanced logistic regression, but improved precision (0.3030) and accuracy (0.7888), indicating a more balanced ability to identify subscribers.

XGBoost achieved the best balance between recall and precision over all models, with a recall of 0.6531, precision = 0.3259, accuracy = 0.8014, and F1-Score = 0.4349. Although its accuracy is lower than the baseline, it represents a substantial improvement in subscriber detection under class imbalance.

Compared with the baseline model, all substantive models improved dramatically (Appendix, Table 11). The baseline achieved 0.883003 accuracy but failed to detect positive subscription term deposit cases. This comparison confirms that accuracy alone is not sufficient in imbalanced classification problems. From a business perspective, models with higher recall reduce the risk of missing potential subscribers, while precision is important to avoid excessive outreach.

While some models achieve higher precision, they fail to identify a large proportion of potential subscribers due to low recall. The XGBoost (Tuned) model provides the most effective trade-off, achieving substantially higher recall while maintaining reasonable precision, making it the most practical solution for marketing campaigns under class imbalance constraints.

6.1.3 ROC Curve Comparison

The ROC curves show moderate discrimination across the evaluated models, with ROC-AUC values ranging from 0.50 to 0.80. XGBoost achieved the highest ROC-AUC (0.8010), followed by Random Forest (0.7867), while both Logistic Regression models presented similar performance, with ROC-AUC values of 0.7731 for the balanced version and 0.7724 for the pipeline model (Appendix Table 11 and Figure 9). This result is relevant because ROC-AUC reflects the model's ability to rank observations across all possible classification thresholds rather than evaluating performance at a single decision point.

The comparison indicates that ensemble methods, particularly XGBoost, provide a slightly stronger overall capacity to distinguish between subscribers and non-subscribers. However, the relatively small differences across models suggest that ranking improvements are incremental, reinforcing the importance of also considering precision and recall when selecting the most appropriate model for an imbalanced business problem (Saito & Rehmsmeier, 2015).

6.2 Reliability

Model reliability was assessed by examining the consistency of performance across multiple evaluation metrics. The results show that model performance varies depending on the metric considered, highlighting the risk of relying on a single indicator such as accuracy.

The DummyClassifier baseline demonstrates minimal reliability, achieving high overall accuracy only by predicting the majority class and completely failing to identify potential subscribers. The Pipeline Logistic Regression shows moderate improvement but remains biased toward the majority class, with low recall undermining its reliability for identifying actual subscribers.

In contrast, the Balanced Logistic Regression, Random Forest, and XGBoost models exhibit more consistent behavior across recall, precision, F1-score, and ROC-AUC. This more balanced performance suggests that these models provide more dependable predictions and are better suited for real-world marketing applications. To complement aggregate performance metrics, an additional error-profile analysis was conducted to identify whether false negatives and false positives were concentrated in specific customer groups.

6.2.1 Profile of False Negatives and False Positives

Error profiling indicated that the final model's mistakes were not uniformly distributed across customer groups. Most errors occurred among Never_Contacted clients, while customers with a prior Success interaction were rarely missed as true subscribers. The unknown contact category showed a higher concentration of false negatives than the other contact methods, and customers with housing loans also showed a higher false-negative share within their error profile.

Operationally, this means that customer segments with limited prior interaction history or incomplete contact information may require additional review or adjusted targeting rules before deployment. These results suggest that the model is less reliable for certain subgroups and that future refinement should focus on reducing missed subscribers in those segments.

6.3 Best Model Selection

Based on the full test-set evaluation, XGBoost is considered the most robust model. It achieved the highest recall (0.6531) and the highest ROC-AUC (0.8010), indicating a stronger overall ability to identify potential subscribers and rank customers according to their likelihood of acceptance. Random Forest showed similar behavior, although with slightly lower performance across all key metrics.

In contrast, DummyClassifier baseline highlights the challenge of class imbalance, achieving high overall accuracy but completely failing to identify potential subscribers. The Pipeline Logistic Regression improved slightly over the baseline in identifying subscribers but remained biased toward the majority class, resulting in low recall (0.1796), meaning that a substantial proportion of potential subscribers would be missed. The Balanced Logistic Regression further improved recall (0.6389), making it more aligned with the business objective of maximizing customer identification, although its overall performance remained slightly below of XGBoost.

These results highlight an important trade-off between precision and recall. In marketing campaigns, false negatives correspond to missed business opportunities, making recall the primary metric of interest. Therefore, XGBoost achieves the most effective balance between predictive performance and business relevance, positioning it as the most suitable model for deployment in this project.

However, this improvement in recall comes at the cost of lower precision, meaning that the model generates a higher number of false positives. This trade-off is acceptable in the context of marketing campaigns, where missing potential subscribers is considered more costly than contacting uninterested customers.

6.4 Cross-validation of Final Model

Cross-validation was performed on the final XGBoost model to assess its reliability and stability across different subsets of the training data. The results (Appendix Table 12) indicate a consistently high recall, with per-fold values ranging from 0.6147 to 0.662 and a mean CV recall of 0.64, confirming the model's effectiveness in identifying a large proportion of potential subscribers.

Furthermore, the precision per fold, ranging from 0.3306 to 0.3376 with a mean of 0.331, suggests that, while the model captures many true positives, it also generates a moderate number of false positives. Overall, these results confirm that XGBoost prioritizes recall over precision, which is aligned with the business objective of minimizing missed subscription opportunities. The relatively low variability across folds indicates that the model is stable and not dependent on a specific train-test split.

6.5 Feature Importance of the selected model

Feature importance analysis highlights the variables most influential in predicting term deposit subscriptions (Appendix Figure 10). `Previous_Interaction_Status_Success` emerges as the top predictor, reflecting the strong impact of prior successful customer engagement. This is followed by `Contact_Method_Unknown`, capturing cases with missing or unspecified contact information, and `Housing_Loan`, which serves as a proxy for the customer's financial profile. The horizontal bar chart illustrates the relative importance of each feature, with longer bars indicating greater influence.

Overall, these results indicate that previous customer interactions and contact-related variables are key drivers of subscription behavior, while financial characteristics also play a meaningful role in the model's decisions.

7. Deployment/Reflection

7.1 Model application

The final model developed in this project can be integrated into the bank's marketing decision making process to support more efficient telemarketing campaigns. Instead of contacting customers randomly or relying solely on historical intuition, the model can estimate the probability that a client will subscribe to a term deposit prior to contact.

These predictions can be used to rank potential clients according to their likelihood of acceptance, allowing marketing teams to prioritize those with the highest predicted probability of subscription.

In practice, this approach would allow financial institutions to allocate their marketing resources more efficiently by focusing efforts on customers with a higher likelihood of accepting the offer. This could reduce unnecessary contact attempts, improve campaign conversion rates, and ultimately enhance the overall efficiency of direct marketing strategies.

By targeting customers more effectively, the bank may also experience higher engagement levels, particularly among high-value segments identified by the model. More personalized marketing offers could lead to increased sales through improved conversion rates, while also enhancing customer satisfaction, as clients are more likely to receive products and services that align with their financial needs and preferences.

From a business perspective, this translates into a clear operational advantage. Targeting high-probability customers reduces unnecessary contact attempts, improves conversion rates, and enhances overall campaign efficiency. At the same time, it enables a more strategic balance between customer acquisition (recall) and cost control (precision).

Additionally, the probability scores generated by the model could be integrated into Customer Relationship Management (CRM) systems to support automated campaign targeting and strategic decision-making. Overall, the model provides a scalable and data driven framework for optimizing marketing campaigns, improving both operational efficiency and business outcomes.

7.2 Limitations

Despite the results obtained in this analysis, several limitations should be acknowledged. First, the dataset represents marketing campaigns conducted by a Portuguese banking institution during a specific time period. As a result, the behavioral patterns captured in the data may not fully generalize to other financial institutions, geographic regions, or economic conditions.

Second, although the models addressed the issue of class imbalance, predicting successful subscriptions remains challenging due to the relatively small proportion of positive outcomes in the dataset. This imbalance may limit the model's ability to consistently identify all potential subscribers without increasing the number of false positives. This reinforces the presence of a trade-off where improving the detection of potential subscribers often leads to an increase in false positives.

Another limitation concerns the availability of information at prediction time. Certain variables in the original dataset, such as call duration, were excluded because they introduce data leakage, ensuring realistic model deployment. However, this also removes variables that may contain strong predictive signals.

Additionally, the “unknown” category is a further limitation in some categorical variables, such as job, education, and contact introducing uncertainty. In these cases, the label represents missing or unrecorded information rather than a true customer characteristic.

Although these values were retained to preserve the original dataset structure and avoid introducing bias through imputation, their presence may reduce the interpretability of the model and limit the precision with which certain customer profiles can be characterized.

Consequently, the predictive power of these variables may be partially constrained by the incomplete nature of the underlying information. Notably, despite being labeled as “unknown”, `Contact_Method_unknown` emerged as one of the most important predictors in the XGBoost model, highlighting the strong informational signal contained even in missing or unspecified contact data.

Finally, the analysis focuses primarily on structured demographic and campaign data. Other potentially relevant factors, such as customer behavioral data, digital interaction history, or macroeconomic indicators, were not included in the dataset and the modeling process, which could provide deeper insights into customer preferences and improve predictive accuracy.

7.3 Improvements

The following improvements are prioritized based on their expected business impact and implementation feasibility. While the current analysis provides a robust modeling framework, several high-impact improvements can be explored to enhance both predictive performance and practical applicability.

First, threshold optimization represents the most immediate and actionable extension of this work. In this project, all models were evaluated using the default classification threshold of 0.5. However, this threshold was not optimized as part of the final modeling pipeline. Adjusting the decision threshold would provide a direct mechanism to control the trade-off between recall and precision, allowing the model to be better aligned with specific business objectives. For example, lowering the threshold would increase recall by identifying more potential subscribers, while increasing it would reduce unnecessary customer contact by improving precision.

Second, more advanced approaches to handling class imbalance could further improve model performance. While this project incorporated class-weight adjustments, additional techniques such as synthetic sampling or more refined cost-sensitive learning strategies could be explored to better capture the minority class without excessively increasing false positives.

Third, model interpretability and business insight generation could be strengthened through the use of explainability techniques. Methods such as SHAP or LIME would allow for a deeper understanding of the factors driving individual predictions, enabling marketing teams to better tailor their communication strategies and improve customer targeting.

Other potential extensions, such as incorporating additional data sources (e.g., behavioral or transactional data) or exploring more complex feature engineering strategies, may also contribute to improved performance. However, the three improvements outlined above represent the most immediate and high-value next steps for enhancing the practical impact of this project.

8. Conclusion

8.1 Final Summary of Findings

This project developed and compared five classification models for predicting whether a client is going to subscribe to a term deposit or not using the Bank Marketing dataset from UCI Machine Learning Repository. The analysis followed a structured workflow that included dataset auditing, exploratory analysis, leakage-aware preprocessing, baseline benchmarking, cross-validated model tuning, and final test-set evaluation.

The dataset was structurally clean, but exploratory analysis showed clear differences in feature scale, and strong positive skew in several variables. These findings justified careful preprocessing and cautious interpretation. Compared with the baseline model, all substantive models performed substantially better on the test set.

8.2 Final Model Recommendation

Based on the overall evaluation (Appendix Table 11), XGBoost is recommended as the final model for this project. It provides the best balance between predictive performance and business relevance, achieving the highest recall and ROC-AUC among all evaluated models. This indicates a strong ability not only to correctly identify potential subscribers but also to effectively rank customers according to their likelihood of accepting the offer.

From a business perspective, the model provides a practical balance between maximizing customer acquisition through higher recall and controlling marketing costs through reasonable precision. While Logistic Regression Balanced achieved comparable recall, XGBoost delivered more consistent overall performance across multiple evaluation metrics, including F1-score and ROC-AUC. This makes it better suited for deployment in a marketing setting where identifying likely subscribers must be balanced against the operational cost of contacting non-subscribers.

Therefore, XGBoost represents the most suitable model for deployment, as it aligns with the objective of improving marketing efficiency while maintaining a practical balance between false positives and false negatives.

8.3 Closing Reflection on Metric Choice in Marketing Field

This project highlights the importance of selecting evaluation metrics that align with business objectives. Although accuracy is commonly reported, it proved to be misleading in this case because of the strong class imbalance in the dataset. For example, the Pipeline Logistic Regression model achieved high accuracy but failed to identify many potential subscribers, resulting in a high number of false negatives.

Error profiling further showed that misclassifications were not evenly distributed across customer groups, suggesting that prior interaction history, contact conditions, and housing loan status influenced where the model was more likely to miss potential subscribers.

In the context of marketing campaigns, recall emerges as a particularly relevant metric because it reflects the model's ability to capture customers who are likely to accept the offer. Missing these customers represents lost business opportunities, making false negatives more costly than false positives. However, recall should not be considered in isolation. A balanced evaluation that also includes precision, F1-score, and ROC-AUC is necessary to ensure that gains in customer detection do not create excessive operational costs. Because the target classes are imbalanced, precision-recall trade-offs are more informative than accuracy alone when evaluating practical subscriber detection performance (Saito & Rehmsmeier, 2015).

Although threshold optimization was not implemented in the final model, it remains an important practical extension. Adjusting the classification threshold would allow organizations to control the balance between recall and precision depending on campaign priorities, making it a valuable tool for real-world deployment.

Overall, this analysis demonstrates that model evaluation in marketing applications must go beyond traditional headline metrics and focus on measures that reflect genuine business impact. Aligning model performance with strategic objectives is essential for developing predictive solutions that are both effective and actionable.

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Appendix

1. Appendix Figures

Figure 1: *Numerical Predictors' Density Distribution*

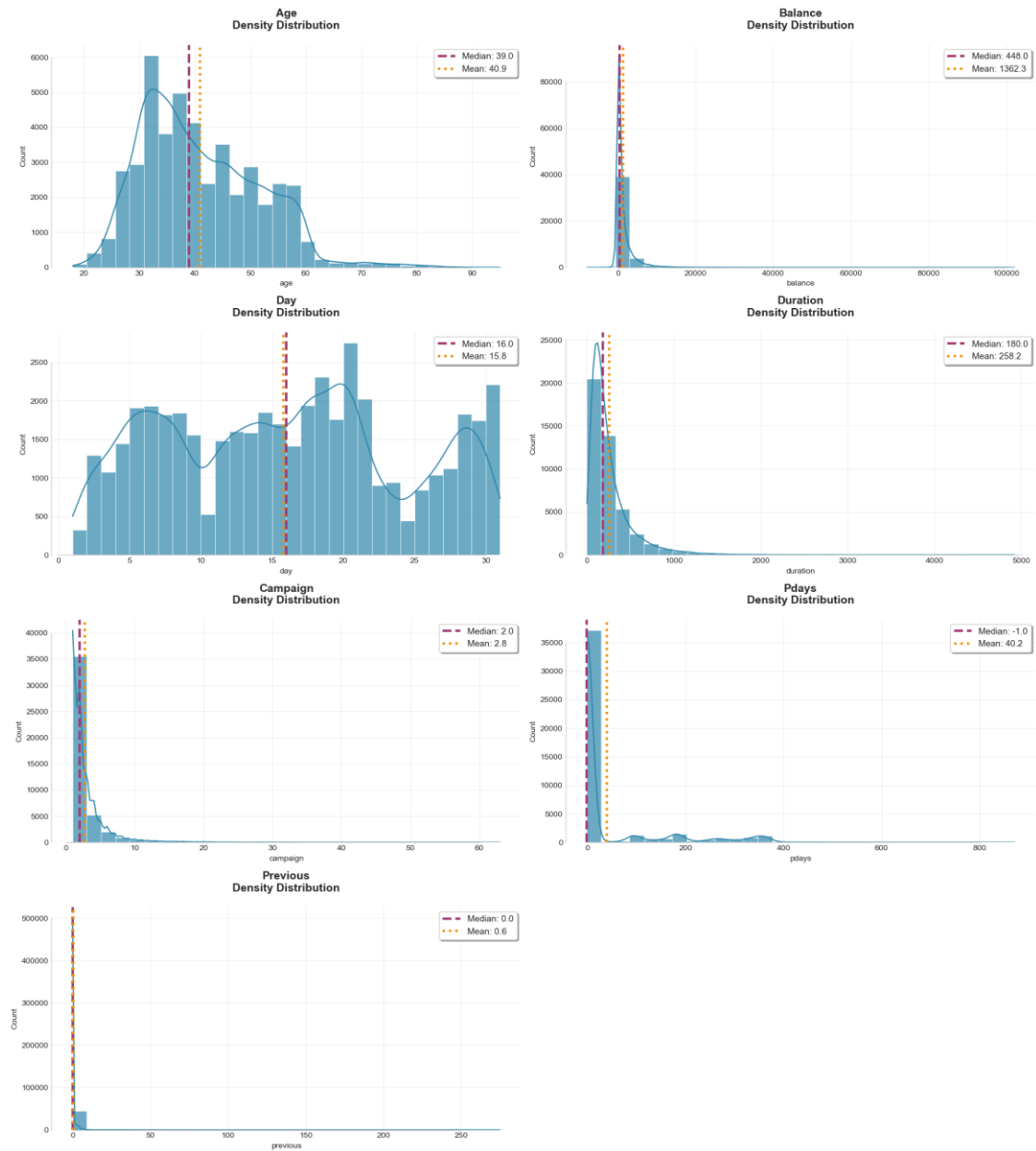


Figure 2. Bar chart Target variable

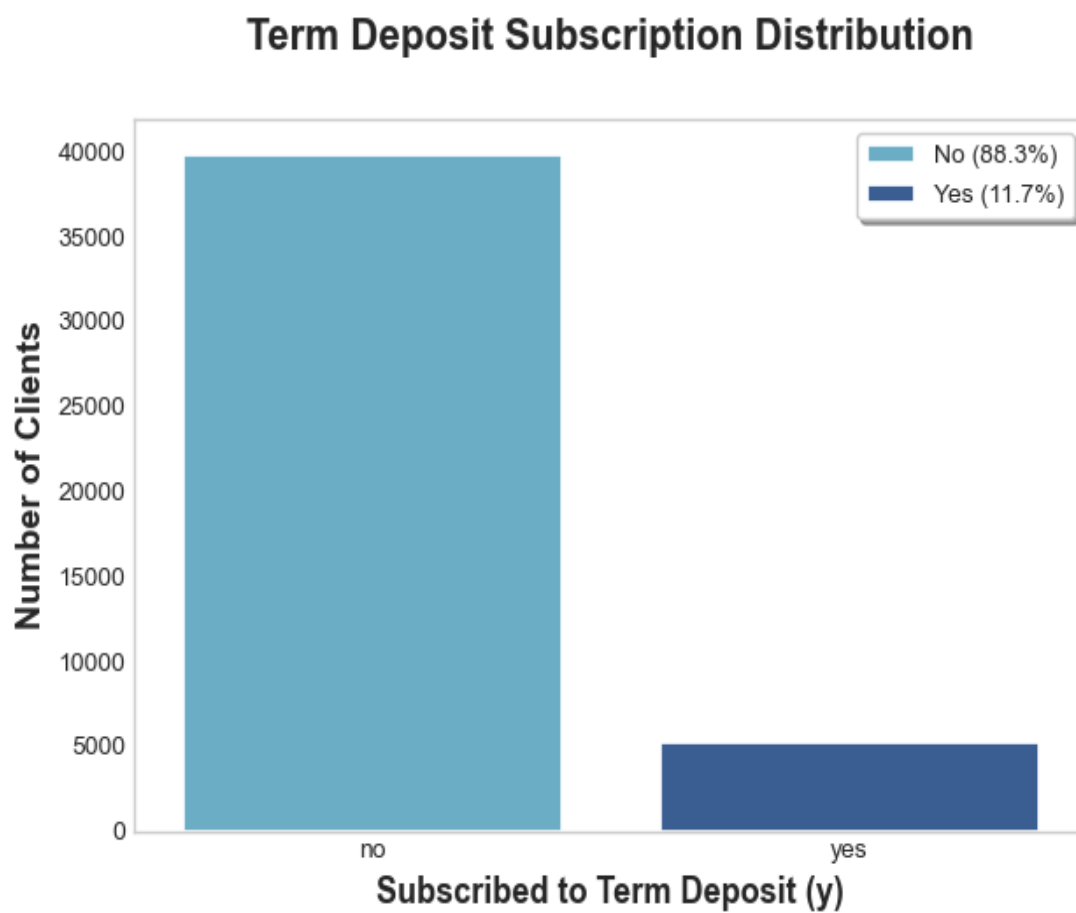


Figure 3. Correlation Matrix of Numeric Predictors

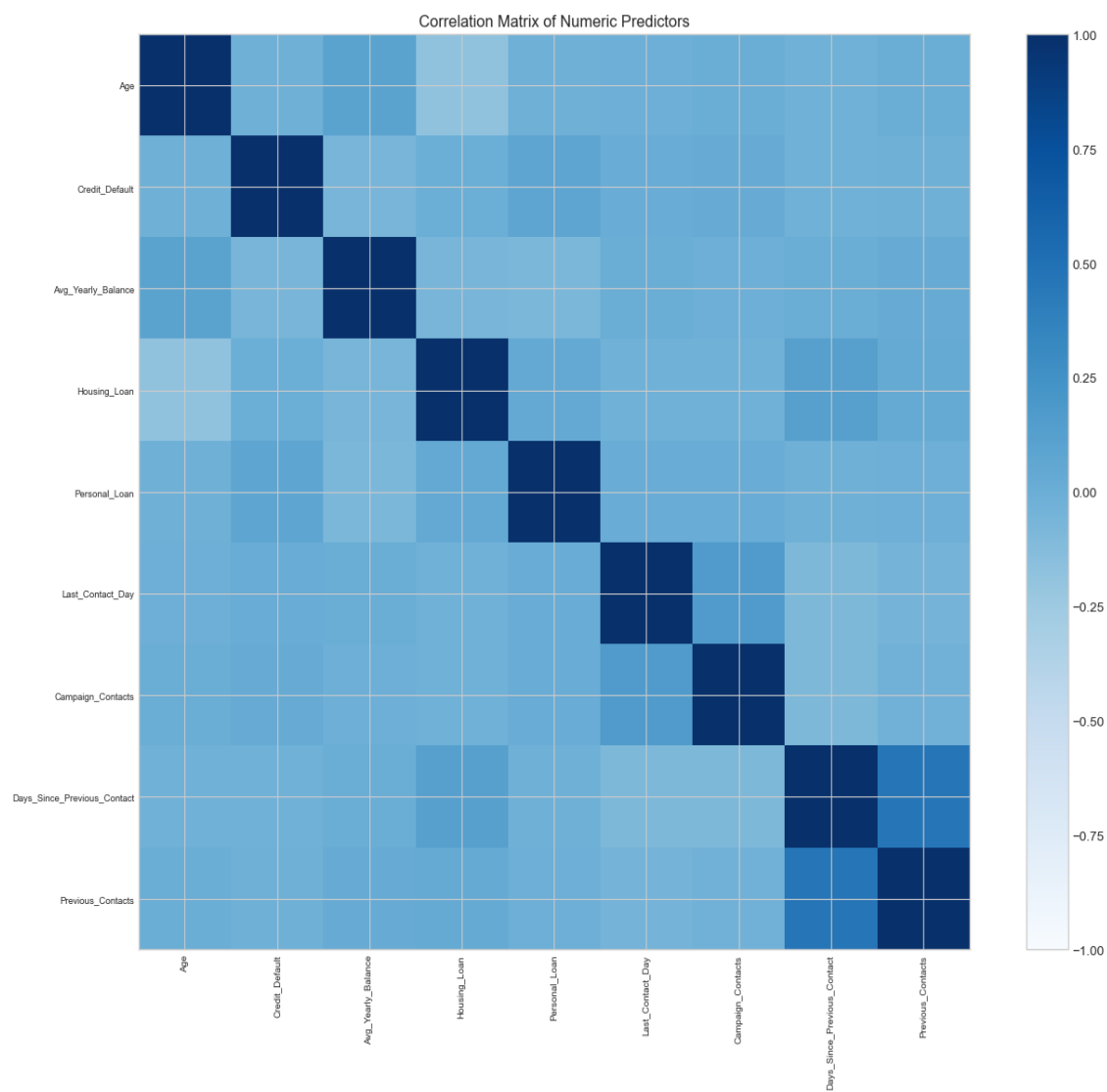


Figure 4. *Confusion Matrices*

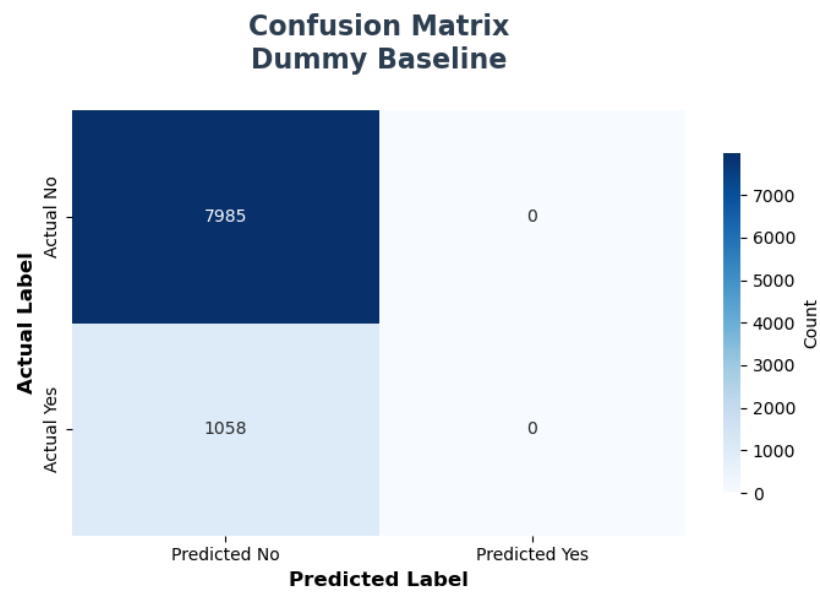


Figure 5. *Confusion Matrix: Logistic Regression (Pipeline) Model*

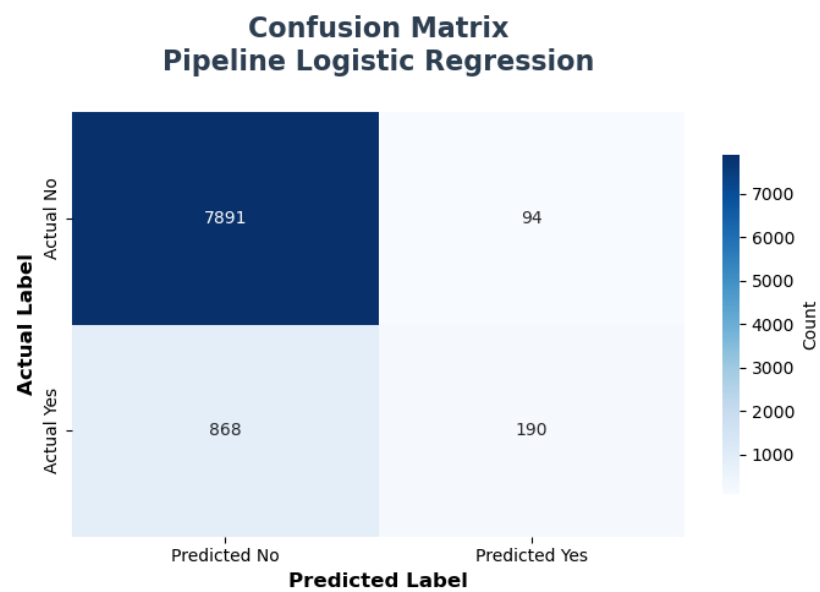


Figure 6. Confusion Matrix: Logistic Regression (Balanced) Model

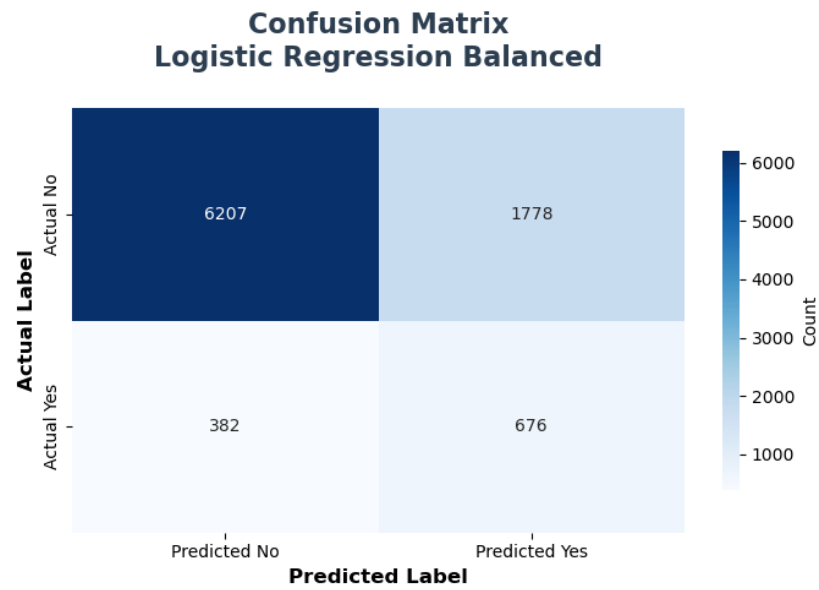


Figure 7. Confusion Matrix: Random Forest Model

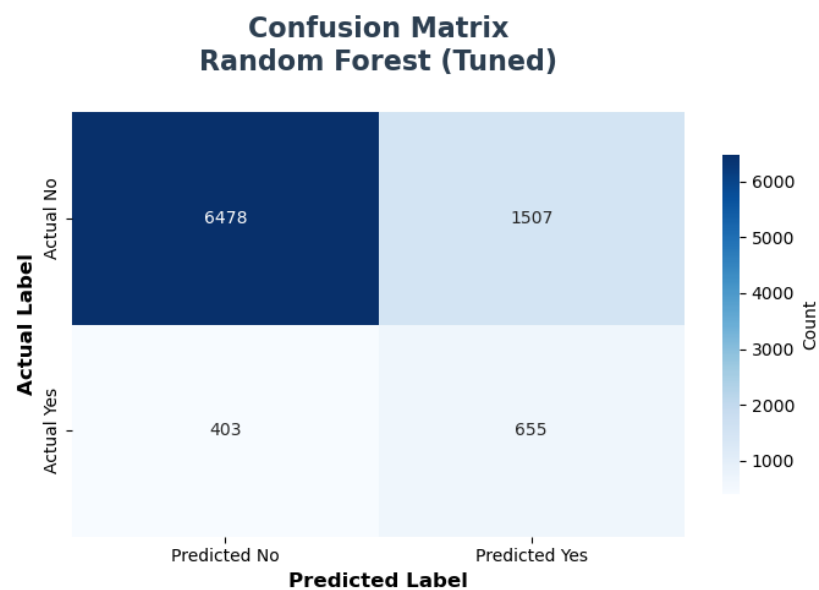


Figure 8. Confusion Matrix: XGBoost Model

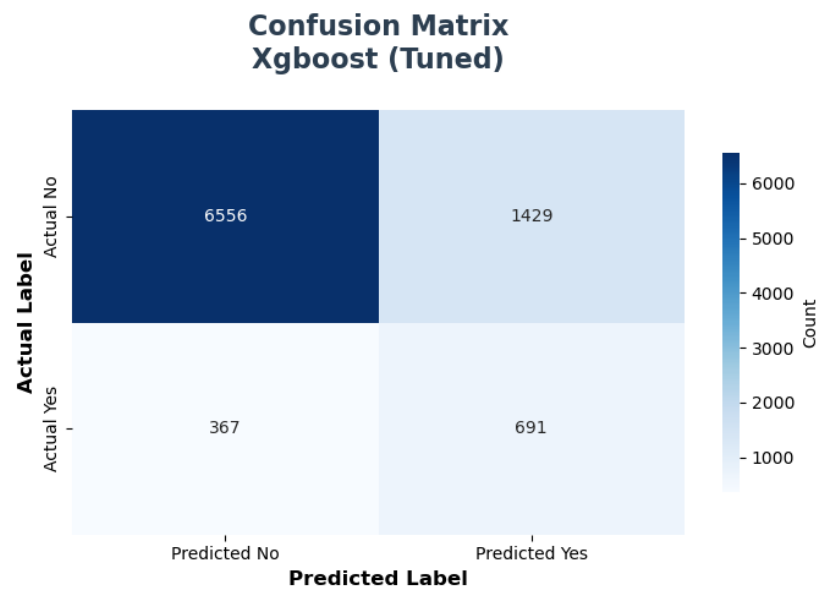


Figure 9. ROC Curve Comparison

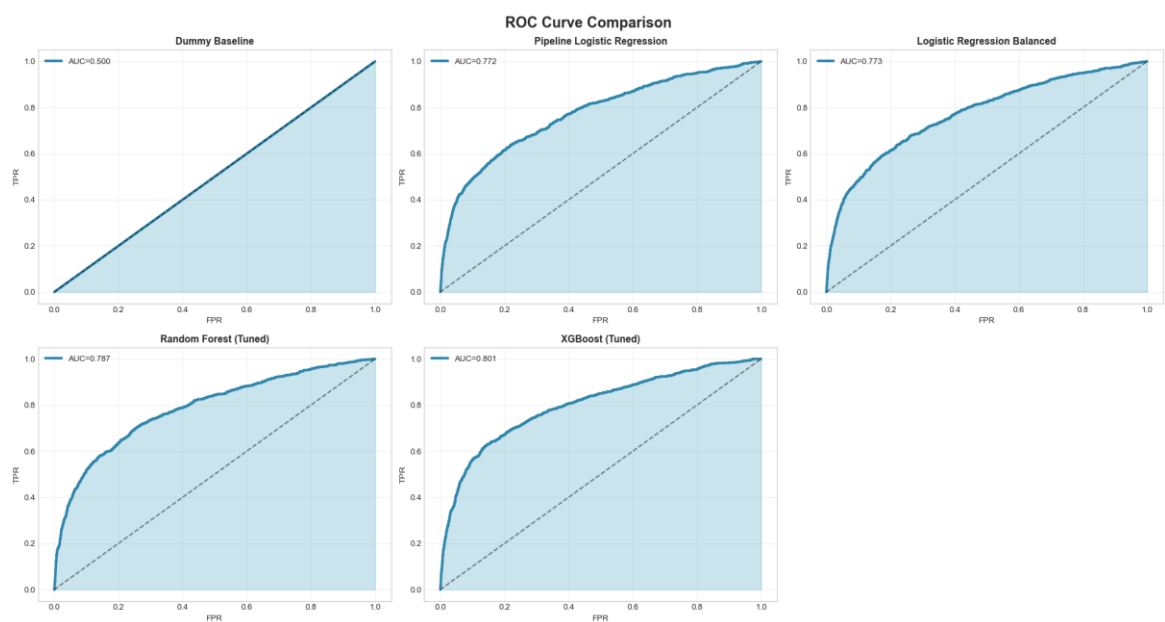
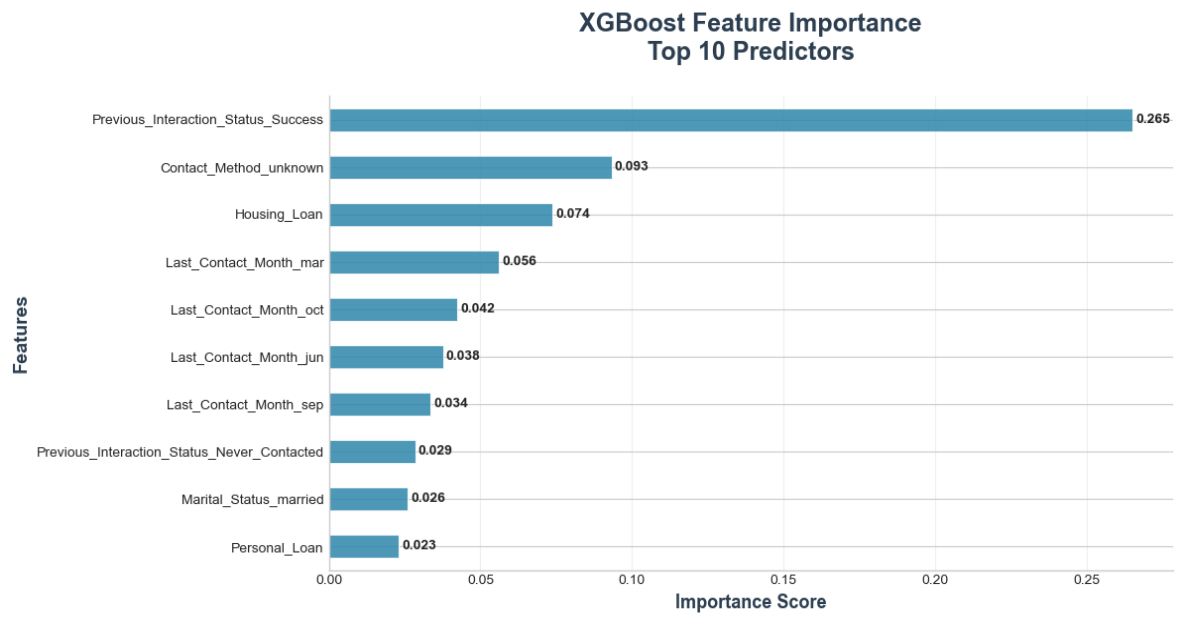


Figure 10. XGBoost Feature Importance



2. Appendix Tables

Table 1. "Unknown" category distribution

Feature	Count
job	288
education	1857
contact	13020
poutcome	36959

Table 2. Descriptive statistics for numeric variables

	mean	std	min	25%	50%	75%	max	range	iqr
age	40.936210	10.618762	18.0	33.0	39.0	48.0	95.0	77.0	15.0
balance	1362.272058	3044.765829	-8019.0	72.0	448.0	1428.0	102127.0	110146.0	1356.0
day	15.806419	8.322476	1.0	8.0	16.0	21.0	31.0	30.0	13.0
duration	258.163080	257.527812	0.0	103.0	180.0	319.0	4918.0	4918.0	216.0
campaign	2.763841	3.098021	1.0	1.0	2.0	3.0	63.0	62.0	2.0
pdays	40.197828	100.128746	-1.0	-1.0	-1.0	-1.0	871.0	872.0	0.0
previous	0.580323	2.303441	0.0	0.0	0.0	0.0	275.0	275.0	0.0

Table 3. Chi-square test for variables with “unknown

Feature	Chi-square result
job	3.337121944935502e-172
education	1.6266562124072994e-51
contact	1.251738325340638e-225
poutcome	0.0

Table 4. Initial skewness screening before feature engineering

Feature	Skewness	Suggested Transformation
age	0.685	Box-Cox
balance	8.360	Yeo-Johnson
duration	3.144	Yeo-Johnson
campaign	4.899	Box-Cox
pdays	2.616	Yeo-Johnson
previous	41.86	Yeo-Johnson

Table 5. Distribution of target variable

y	Count	Proportion
No	39,922	0.883
Yes	5,289	0.117

Table 6. Outlier analysis

	Column	Z-Score Outliers	IQR Outliers
0	age	381	487
1	balance	745	4729
2	day	0	0
3	campaign	840	3064
4	pdays	1723	8257
5	previous	582	8257

Table 7. Top 15 strongest absolute correlations

	feature_1	feature_2	correlation	abs_corr
35	Days_Since_Previous_Contact	Previous_Contacts	0.454820	0.454820
2	Age	Housing_Loan	-0.185513	0.185513
30	Last_Contact_Day	Campaign_Contacts	0.162490	0.162490
24	Housing_Loan	Days_Since_Previous_Contact	0.124178	0.124178
1	Age	Avg_Yearly_Balance	0.097783	0.097783
31	Last_Contact_Day	Days_Since_Previous_Contact	-0.093044	0.093044
33	Campaign_Contacts	Days_Since_Previous_Contact	-0.088628	0.088628
16	Avg_Yearly_Balance	Personal_Loan	-0.084350	0.084350
10	Credit_Default	Personal_Loan	0.077234	0.077234
15	Avg_Yearly_Balance	Housing_Loan	-0.068768	0.068768
8	Credit_Default	Avg_Yearly_Balance	-0.066745	0.066745
32	Last_Contact_Day	Previous_Contacts	-0.051710	0.051710
21	Housing_Loan	Personal_Loan	0.041323	0.041323
25	Housing_Loan	Previous_Contacts	0.037076	0.037076
34	Campaign_Contacts	Previous_Contacts	-0.032855	0.032855

Table 8. *Original vs Enhanced Column Names*

Original	Enhanced Name
age	Age
job	Occupation
marital	Marital_Status
education	Education_Level
default	Credit_Default
balance	Avg_Yearly_Balance
housing	Housing_Loan
loan	Personal_Loan
contact	Contact_Method
day	Last_Contact_Day
month	Last_Contact_Month
duration	Duration
campaign	Campaign_Contacts
pdays	Days_Since_PreviousContact
previous	Previous_Contacts
poutcome	Previous_Campaign_Outcome
y	Sub_Term_Deposit

Table 9. Top 10 multicollinearity results

Feature	VIF Score
Previous_Interaction_Status_Never_Contacted	8.8511
Last_Contact_Month_may	5.4251
Marital_Status_married	5.4142
Education_Level_secondary	4.3381
Education_Level_tertiary	4.2711
Occupation_management	3.5355
Marital_Status_single	3.5307
Contact_Method_unknown	3.3232
Last_Contact_Month_jun	3.2473
Housing_Loan	3.0109

Table 10. Class proportion before and after splitting

Training class summary:		
	count	proportion
Sub_Term_Deposit		
0	31937	0.883
1	4231	0.117
Test class summary:		
	count	proportion
Sub_Term_Deposit		
0	7985	0.883
1	1058	0.117

Table 11. Model Comparison

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
4	XGBoost (Tuned)	0.801393	0.325943	0.653119	0.434865	0.800951
2	Logistic Regression Balanced	0.761141	0.275469	0.638941	0.384966	0.773086
3	Random Forest (Tuned)	0.788787	0.302960	0.619093	0.406832	0.786737
1	Pipeline Logistic Regression	0.893619	0.669014	0.179584	0.283159	0.772415
0	Dummy Baseline	0.883003	0.000000	0.000000	0.000000	0.500000

Table 12. Cross-Validation Results

Metric	Recall per fold	Mean CV
Recall	[0.66193853 0.64775414 0.64817001 0.62529551 0.61702128]	0.6400358936142301
Precision	[0.33755274 0.32834032 0.32180539 0.33587302 0.33058898]	0.3308320910269881